

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, AMT)

January 2014

ME – 734 Computer Aided Engineering

Date: 06.01.2014, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION- I

- Q1 (a) Differentiate between classical design & computer aided design process. 5
- (b) Explain parametric continuity conditions for curves. 3
- (c) Explain most common mating conditions for locating and orienting parts in their assembly. 3
- Q2 (a) Compare the capabilities of types of displays. 3
- (b) Explain the working principle of digitizer. 3
- (c) Using Bresenham's line algorithm, find the pixel positions between end points (10, 5) and (15, 9). 6

OR

- Q2 (a) What is scan conversion? Why can't a vector scan display produce as many colours as a raster scan display? 3
- (b) Differentiate between Analytic curves and Parametric curves. 3
- (c) How can you generate a circle with centre 'c' and radius 'r' using Bresenham's algorithm. 6
- Q3 (a) Compare geometric transformation with coordinate transformation. Also explain composite transformation. 3
- (b) What is homogenous coordinate system? Why is it preferred in computer graphics? 3
- (c) A square having end points A(1,1), B(6,1), C(6,6), D(1,6) is rotated by 50° clockwise direction keeping point (6,1) fixed. Find the final coordinates. 6

OR

- Q3 (a) Write 3-D transformation matrix for rotation about x, y, z-axis. 3
- (b) Prove that 3D rotations are not commutative. 3
- (c) Find the reflection of point (5, 5) about a line $y = 3x + 8$. 6

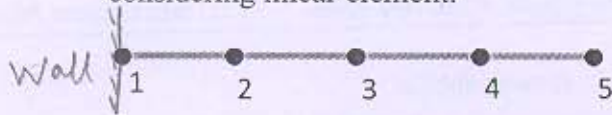
SECTION- II

- Q4 (a) What is reverse engineering? Justify its use. 4
- (b) Explain the principle of Rapid Prototyping its advantages and applications. 3
- (c) Classify Rapid Prototyping processes and explain any one of them. 3

- Q5 In a 1D finite element with two nodes and one temperature degree of freedom at each of the node, derive shape function matrix $[N]$ by assuming suitable temperature function. 7

OR

- Q5 A fin of the length 44 mm is divided in to four equal parts as shown below to analyze the heat transfer. It has five nodes 1, 2, 3, 4 and 5 having temperature 130, 150, 300, 310 and 315°C, respectively. Find the temperature at point 20 mm and 40 mm from the wall by considering linear element. 7



- Q6 Attempt any *three*: 18

- (a) Derive the finite difference formula to compute $d\phi/dx$ using polynomial approach.
- (b) A triangular element has nodes at (2, 5), (7, 6) and (6, 10) coordinates having temperature 130°C, 260°C and 390°C respectively. Determine temperature at point (6, 7).
- (c) Explain the explicit and implicit approach for temporal discretization with its advantages and limitations
- (e) Explain Jacobi method for solving system of simultaneous equation and enlist the steps involved in solution algorithm.
- (f) Discuss mathematical and numerical modeling of any one manufacturing process.
